

Problem Set 7
COMP301 Spring 2025
Week 8: 07.04.2025 - 11.04.2025

Instructions:

- Submit your answers to the Blackboard PS7 assignment until 12.04.2024 Saturday, at 23.59.
- Submit your code and PDF file to LearnHub as a single zip file *yourIDno_username.zip*.
- Read the questions carefully. Good luck!

Problem 1: ¹ Extend the Let language by adding the following new operations `minus` and `prime?`. You need to work with Let language codes for this problem.

Hint: You only need to change `lang.rkt` and `interp.rkt`.

Hint: For part B you can write a helper functions for calculating the prime number checker.

Part A. Extend the Let language by adding a new operation `minus` that takes one argument `n` and returns `-n`. For example, the value of `minus(-(minus(5), 9))` should be 14, which is equal to `-((-5) - 9)`.

Part B. In this part, you will add the `prime?` operation to the Let language. The `prime?` will check if the given input integer `x` can be divisible by a smaller number than input bigger than zero. For example, `prime?(2)` will return output `#t`, whereas `prime?(4)` returns output `#f`

After you finish implementing the described operations, uncomment and run the tests in `tests.rkt`. At the end of that file we defined extra tests for `minus` and `prime?` you can use this as an example to how to write test cases and define your own tests.

Problem 2: Fill in the blanks for the output of this interpreter. Assume $\rho_0 = [x = 20, y = 15, z = 10]$. Note that below snippet is not complete, although we only want answers to the `_____`'s, we recommend you to continue the evaluation as a self study exercise. Please also include the output of the program in your answer.

(Please note that ρ_0 denotes the initial environment. The notation $[20]$ indicates an EXPVAL type with the value 20, not a Scheme integer. To demonstrate environment extension, please follow the notation used in the following example: $n = [13] \rho$.)

```
let n = 20 in if zero?(-(x, n)) then -(x, y) else -(-(z, n), -(y, x))
(value-of
  <<let n = 15
    in if zero?(-(x, n)) then
      -(x, y)
    else -(-(z, n), -(y, x))>>
  ρ0)

= (value-of
  <<if zero?(-(x, n)) then
    -(x, y)
  else -(-(z, n), -(y, x))>>
```

¹EOPL p.72 Exercise 3.6

____[1]____)

Let $\rho_1 =$ ____[2]____

= (**if** (expval->bool (value-of <<**zero?**(-(x, n))>> ρ_1))
 (value-of <<-(x, y)>> ρ_1)
 (value-of <<-(-(z, n), -(y, x))>> ρ_1))

= (**if** ____[3]____
 (value-of <<-(x, y)>> ρ_1)
 (value-of <<-(-(z, n), -(y, x))>> ρ_1))

= (value-of <<____[4]____>> ρ_1)

= [(-
 [(value-of <<____[5]____>> ρ_1)]
 [(value-of <<____[6]____>> ρ_1)])]

= ____[7]____